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CUSTOMIZABLE INTERCONNECTING BUILDING BLOCKS

Abstract

The present invention provides customizable building blocks comprising two “half” blocks that may be interconnected to form a “whole” block, allowing for the half blocks to be rotated and reattached to one another, thus changing the configuration and orientation of male and female connectors on the outer faces of the whole blocks. In a preferred embodiment, two half blocks combine to form a whole block, with each half block having an inner face, and outer face, and a plurality of side faces. The orientation of each half block when conjoined with another half block determines the number and orientation of male and female connectors on the outer faces of a whole block. The feature of conjoining two half blocks to create a whole block allows for the male and female connectors to be moved to different faces of the block depending on the desired building function and orientation.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application No. 63/561,807, filed on Mar. 6, 2024, titled “CUSTOMIZABLE INTERCONNECTING BUILDING BLOCKS”, the contents of which are incorporate herein by reference.

FIELD OF THE INVENTION

[0002] The present invention relates generally to interconnecting building blocks and more specifically to customizable interconnecting building blocks that are adaptable and reconfigurable in order to allow flexible interconnection between multiple blocks.

BACKGROUND

[0003] Interconnecting blocks are well known in the toy industry. Lego® and Mega Bloks® provide two such examples of toy building blocks having integral features that provide for ready interconnection between blocks. A variety of interconnecting blocks are commercially available, and the majority of such blocks operate on the principal that a first block will have a projecting male feature, a second block will have a female feature, and the male feature of the first block will mate to the female feature of the second block to create a semi-permanent mechanical connection between the two. Generally, the mechanical connection will lock the blocks into relative positions with one another, and the blocks cannot be rearranged or re-oriented with respect to one-another without fully disconnecting the blocks and then reconnecting in the new desired configuration.

[0004] The basic form of this type of block generally provides male features on the top surface of a block and female features on the bottom surface of the block. The interconnection between male and female blocks is generally quantized, such that two given blocks are connectible in a finite number of configurations.

[0005] This general design concept has significant implications to the types of structures that can be constructed of these blocks. For example, if a regular rectangular vertical wall is built of basic Lego®-type bricks, any bricks not at the periphery of the wall are fully locked into place and cannot be moved or removed without some degree of disassembly of the entire wall. Furthermore, a wall constructed in such a manner will not have interconnecting features on any faces other than the top and bottom faces. Therefore, it would not be possible to further interconnect additional blocks to the existing wall in any areas other than the top or bottom.

[0006] Specialized interconnecting blocks exist that have multiple faces with male or female interconnection features, but they are not the norm. Typically, significant planning is required to construct a desired structure out of interconnecting toy blocks because of the limited ways in which they can be connected and because of the difficulty of modifying a structure on an ad hoc basis.

[0007] In general, interconnecting building blocks are not reconfigurable, and a given block can only be connected to other blocks in the manners permitted by the existing permanent interconnection feature arrangement. If a given block is not suitable for a given situation or use, then a different, suitable block must be obtained.

[0008] A need exists for an interconnecting block and interconnecting block system that is customizable and reconfigurable and will allow users to connect blocks in a much more open-ended and unconstrained manner.

SUMMARY

[0009] The disclosure presented herein relates to a customizable and reconfigurable interconnecting building block. In some embodiments, the preceding and following embodiments and descriptions are for illustrative purposes only and are not intended to limit the scope of this

disclosure. Other aspects and advantages of this disclosure will become apparent from the following detailed description.

[0010] Certain terminology and derivations thereof may be used in the following description for convenience in reference only, and will not be limiting. For example, words such as “upward,” “downward,” “left,” and “right” would refer to directions in the drawings to which reference is made unless otherwise stated. Similarly, words such as “inward” and “outward” would refer to directions toward and away from, respectively, the geometric center of a device or area and designated parts thereof. References in the singular tense include the plural, and vice versa, unless otherwise noted.

[0011] Existing interconnecting building blocks are generally solid and generally rigid bodies. Each individual building block is manufactured with a given set of interconnection features, and a user is limited to connecting the blocks in manners that the manufactured designs permit. Additionally, the design of the interconnection features generally do not permit customizable or adjustable mechanical connection between the blocks. If a user desires to modify the connection between the blocks, a user is generally required to disconnect the blocks, then reconnect in the new, desired connection.

[0012] The present invention aims to solve the problem of limited modes of interconnection between blocks. The building block of the present invention is reconfigurable to allow a user to construct and reconfigure each individual face to provide male and/or female connecting features to interconnect with an adjacent face of another block. Further, unlike conventional building blocks, the blocks of the present invention are combinable on all sides and faces for a great multitude of combinations.

[0013] One or more embodiments of the present invention provide customizable building blocks comprising two “half” blocks that may be conjoined to form a “whole” block. This feature of a “two-part” block allows for the half blocks to be rotated or reoriented and reattached to one another, thus changing the configuration and orientation of male and female connectors on the outer faces of the whole blocks. In an exemplary shape, the block is a six-sided cube comprised of two conjoined square prisms. In other variations, the shape may be a triangular prism, a half-cylinder, or any other suitable or desired shape while still maintaining the reconfigurability features of the block system.

[0014] In a preferred embodiment, two half blocks combine to form a whole block, with each half block having an inner face, and outer face, and a plurality of side faces. For example, in the case of a cube shaped whole block, the square prism half block would have an inner face, an outer face, and four side faces. And, when the inner face of half block square prism “A” is interconnected with the inner face of half block square prism “B”, it creates a whole block cube with six outer faces.

[0015] It is not necessary, however, to connect two half blocks together to create a whole block. Alternatively, a half block may be used on its own and added to an existing block structure where only a half block is needed or desired.

[0016] The block faces may include a plurality of female and male connectors. In one or more embodiments, the half block outer face has a plurality of female connectors adapted to receive a male connector and, in some embodiments, may also include at least one male connector. In alternative embodiments, the outer face may comprise only of male connectors. The inner face preferably has a plurality of female connectors and at least one male connector. In alternative embodiments, the inner face may be all male connectors, all female connectors, or a plurality of male connectors and at least one female connector. The male connector(s) of the inner face is adapted for interconnection with a corresponding female connector of the inner or outer face of another half block, thereby forming a whole block.

[0017] The side faces include a plurality of female connectors, and at least one side face includes at least one male connector. The orientation of each half block when conjoined with another half block determines the number and orientation of male connectors on the outer faces of a whole

block. The feature of conjoining two half blocks to create a whole block allows for the male connectors to be moved to different faces of the whole block depending on the desired building function and orientation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] Any dimensions expressed or implied in the drawings and these descriptions are provided for exemplary purposes. Thus, not all embodiments within the scope of the drawings and these descriptions are made according to such exemplary dimensions. The drawings are not made necessarily to scale. Thus, not all embodiments within the scope of the drawings and these descriptions are made according to the apparent scale of the drawings with regard to relative dimensions in the drawings.

[0019] The embodiments illustrated, described, and discussed herein are illustrative of the present invention. As these embodiments of the present invention are described with reference to illustrations, various modifications or adaptations of the methods and or specific structures described may become apparent to those skilled in the art. It will be appreciated that modifications and variations are covered by the above teachings and within the scope of the appended claims without departing from the spirit and intended scope thereof. All such modifications, adaptations, or variations that rely upon the teachings of the present invention, and through which these teachings have advanced the art, are considered to be within the spirit and scope of the present invention. Hence, these descriptions and drawings should not be considered in a limiting sense, as it is understood that the present invention is in no way limited to only the embodiments illustrated.

[0020] FIG. 1A is a perspective view of an embodiment of a half block illustrating an outer face and two side faces.

[0021] FIG. 1B is a perspective view of FIG. 1A flipped vertically so that the inner face is viewable.

[0022] FIG. 2A is a perspective view of an embodiment of a half block illustrating an inner face and two side faces wherein one side face lacks female or male connectors.

[0023] FIG. 2B is a perspective view of an embodiment of a half block illustrating an inner face and two side faces wherein one of the side faces has two male connectors.

[0024] FIG. 3 is a perspective view of an embodiment of a whole block created when FIG. 1A and FIG. 1B are conjoined, illustrating three outer faces with only one outer face having male connectors.

[0025] FIG. 4 is a perspective view of FIG. 3 with the top half block rotated 180 degrees to show a whole block illustrating an outer face with a plurality of female connectors and two male connectors on opposing outer faces.

[0026] FIG. 5 is a perspective view of an embodiment of a whole block whereby an inner face is connected to an outer face, resulting in three outer faces with male connectors.

[0027] FIG. 6 is a perspective view illustrating an alternative embodiment of FIG. 5 with the half block on top rotated 90 degrees counterclockwise such that the male connectors are 180 degrees apart.

[0028] FIG. 7 is a perspective view illustrating two blocks of FIG. 3 interconnected so the male connectors are hidden.

[0029] FIG. 8 is a perspective view of the two whole blocks of FIG. 7 conjoined in a chosen orientation such that no male connectors are exposed on the outer faces.

[0030] FIG. 9 is a perspective view of an embodiment of a half block illustrating a shape variation with a rounded edge and female connectors at 45 degrees.

[0031] FIG. 10 is a perspective view of an embodiment of a half block illustrating a shape variation

of a triangular prism.

[0032] FIG. **11** is a perspective view of an embodiment of a half block illustrating a shape variation of a hexagon.

[0033] FIG. **12** is a perspective view of a square prism half block illustrating an alternative embodiment of the male connector.

[0034] FIG. **13** is a perspective view of another embodiment of a whole block illustrating an alternative embodiment of the male and female connectors.

[0035] FIG. **14** is a perspective view of an alternative embodiment of a square prism half block illustrating the inner face.

[0036] FIG. **15** is a perspective view of a whole block illustrating the alternative embodiment of FIG. **14** conjoined with a second half block.

[0037] FIG. **16** is a perspective view of two square prism half blocks illustrating yet another embodiment of the male and female connectors, whereby the two half blocks are aligned to be conjoined.

[0038] FIG. **17A** is a perspective view of two square prism half blocks illustrating an embodiment of block whereby the inner face has different male and female connectors from the outer face and side faces.

[0039] FIG. **17B** is a perspective view of an embodiment of a half block of FIG. **17A**, illustrating an alternative arrangement of the male and female connectors.

[0040] FIG. **17C** is a perspective view of an embodiment of a half block of FIG. **17A**, illustrating an alternative arrangement of the male and female connectors.

[0041] FIG. **18** is a perspective view of an alternative embodiment of a whole block created when the half blocks of FIG. **17A** are conjoined, illustrating only one outer face having male connectors.

[0042] FIG. **19** is a perspective view of an embodiment of a whole block constructed when the top half block from FIG. **18** is rotated 180 degrees, resulting in male connectors on opposing outer faces.

[0043] FIG. **20** is a perspective view of an alternative embodiment of a whole block when the bottom half block from FIG. **18** is rotated 90 degrees clockwise.

[0044] FIG. **21** is a perspective view of an alternative embodiment of a whole block when the top half block from FIG. **19** is rotated 270 degrees clockwise.

[0045] FIG. **22** is a perspective view of two half blocks whereby the female connectors and male connectors are arranged for a staggered interconnection.

[0046] FIG. **23** is a perspective view of an embodiment of a whole block created when the half blocks of FIG. **22** are conjoined, illustrating a staggered interconnection.

[0047] FIG. **24** is a perspective view of two half block illustrating an alternative embodiment of the male connector and female connector whereby the two half blocks can rotate with respect to one another.

[0048] FIG. **25** is a perspective view of an embodiment of a whole block created when the half blocks of FIG. **24** are connected at an offset.

[0049] FIG. **26** is a perspective view of an alternative embodiment of two half blocks aligned to be conjoined.

[0050] FIG. **27** is a perspective view of a whole block illustrating a straight pull injection molding embodiment of the female connector, whereby the female connector is channel-shaped.

[0051] FIG. **28** is a perspective view of an alternative embodiment of two half blocks aligned to be conjoined.

[0052] FIG. **29** is a perspective view of an alternative embodiment of two half blocks aligned to be conjoined.

[0053] FIG. **30** is a perspective view of a plurality of blocks of FIG. **1A** interconnected to illustrate a multitude of building combinations.

[0054] FIG. **31** is a perspective view of a plurality of blocks of an FIG. **17A** interconnected to

illustrate a multitude of building combinations.

[0055] FIG. 32 is an exploded view of a plurality of twelve blocks of FIG. 3 and one block from FIG. 4 aligned for sturdy interconnection

[0056] FIG. 33 is an exploded view of a plurality of thirteen blocks from FIG. 3 aligned for a looser interconnection, wherein two blocks (upper right) will remain unconnected.

[0057] FIG. 34 is a perspective view of FIG. 32 or FIG. 33 assembled, illustrating no male connectors on the side surfaces.

[0058] FIG. 35 is an exploded view of a plurality of blocks from FIG. 4 aligned for interconnection.

[0059] FIG. 36 is a perspective view of FIG. 35 assembled, illustrating a plurality of male connectors on the side surfaces.

DETAILED DESCRIPTION

[0060] The terms used in this specification generally have their ordinary meanings in the art, within the context of the disclosure, and in the specific context where each term is used. Certain terms that are used to describe the disclosure are discussed below, or elsewhere in the specification, to provide additional guidance to the practitioner regarding the description of the disclosure. It will be appreciated that the same thing can be said in more than one way.

[0061] In one or more embodiments, two half blocks 2 are removably conjoined or interconnected with one another to form a whole block 3, with each half block 2 having an inner face 4, and outer face 5, and a plurality of side faces 6. With regard to a whole block 3, all the faces will be referred to herein as outer faces 5.

[0062] The faces of the block include a plurality of female connectors 7 such as cavities, channels, grooves, receptacles, and a plurality of male connectors 8 comprising geometric projections such as tabs, pegs, or the like. The female 7 and male connectors 8 are dimensioned such that the male connector 8 is adapted to fit within a corresponding female connector 7 to allow interconnection of the block faces 4, 5, 6. Although the shape and dimensions of the connectors 7, 8 may vary within the block system 1, this interlocking compatibility remains.

[0063] In one or more embodiments, the unique shape combination of grooves or channel-shaped female connectors 7 and peg-shaped male connectors 8 allows the male connector 8 to be inserted or interconnected at any point along the groove of the female connector 7, thereby allowing an offset or staggered point of interconnect typically not possible with conventional building blocks. In some embodiments, the shape and design of the female connector 7, when interconnected with a corresponding male connector 8, provides rotational movement of one block 2, 3 with respect to another block 2, 3.

[0064] More specifically, in one or more embodiments, the female connector 7 has a length, a width, and a depth; or in the case of a cylindrical-shaped female connector 7, a circumference and a depth. The length of the female connector 7 may be defined as the longitudinal measurement of the female connector 7 from one end to another, while the width may be defined as the shorter measurement of the width between the longitudinal edges of the female connector 7 (i.e. side to side). The depth of the female connector 7 is the depth from the outer surface of the block face 4, 5, 6 inward towards the center of the block. The male connector 8 has a length, a width, and a height; or in the case of a cylindrical-shaped male connector 8, a circumference and a height.

[0065] The dimensions of the male 8 and female connectors 7 are adapted for removable interconnection. The height of the male connector 8 preferably does not exceed the depth of the female connector 7 to allow for full insertion of the male connector 8 of one block face 4, 5, 6, into the female connector 7 of a second block face 4, 5, 6. The width of the female connector 7 preferably corresponds to the dimensions of the male connector 8 such that the male connector 8 can be smoothly inserted into the female connector 7 yet maintain enough frictional force sufficient to hold the two interconnected faces 4, 5, 6 together, such as press fit or interference fit. for example. The length of the female connector 7 may vary and does not need to correspond with the

length, width, or diameter of the male connector **8**. In some cases, the length of the female connector **7** may exceed the length, width, or diameter of the male connector **8**, thereby allowing the male connector **8** to be inserted or interconnected with the female connector **7** at any point along the length of the female connector **7**, as in FIG. 27. This feature allows a first block face **4, 5, 6** to be interconnected with a second block face **4, 5, 6** at an offset and even provides the ability of one block face **4, 5, 6** to interconnect and slide along the length of, or rotate with respect to, another block face **4, 5, 6**, as seen in FIG. 25.

[0066] The female **7** and male connectors **8** may be dimensioned such that the placement of a single connector (male **8** or female **7**) occupies 25 percent or less of the surface area of a block face **4, 5, 6**, thereby allowing a plurality of at least four or more connectors on a single block face **4, 5, 6**. For example, the block of FIG. 1A has four connectors **7** on an outer face **5**, the block of FIG. 11 has twelve connectors **7,8** on an inner face **4**, and the block of FIG. 22 has eight connectors **7,8** on an inner face **4**. A greater number of connectors per block and per block face **4, 5, 6** can provide more versatility in block orientation and assembly.

[0067] The half block **2** outer face **5**, as shown in FIG. 1A for example, preferably has a plurality of female connectors **7** adapted to receive a male connector **8**. In some embodiments, as shown in FIG. 2A for example, the outer face **5** may also include at least one male connector **8**. In alternative embodiments, the half block **2** outer face **5** may comprise a plurality of male connectors **8** and no female connectors **7** (FIG. 5).

[0068] The inner face **4** preferably has a plurality of female connectors **7** and at least one male connector **8**, such as FIG. 2B. In a preferred embodiment, the male connector(s) **8** of the inner face **4** is adapted for interconnection with a corresponding female connector **7** of an inner face **4**, outer face **5**, or side face **6** of another half **2** or whole block **3**. In this way, half **2** and whole blocks **3** may be assembled side face **6** to side face **6**, inner face **4** to inner face **4**, inner face **4** to outer face **5**, outer face **5** to outer face **5**, side face **6** to either inner **4** or outer face **5**, etc. The block system **1** of the present invention includes combinable features on all faces **4, 5, 6**, of the block, thereby allowing for a multitude of interconnection capabilities and combinations, as illustrated in FIGS. 30-36.

[0069] For example, if a rectangular vertical wall is built of basic Lego-type bricks, any bricks not at the periphery of the wall will not have interconnecting features on any faces other than the top and bottom faces. Therefore, it would not be possible to further interconnect additional blocks to the existing wall in any areas other than the top or bottom. The blocks of the present invention, however, include male **8** and female connectors **7** on all faces **4, 5, 6** of the block **3** that may be reconfigured or re-oriented with respect to one another. A rectangular vertical wall built with the blocks of the present invention have the capability of including male **8** and female connectors **8** on the sides of the wall (See FIGS. 30 and 31), not just top and bottom. This feature allows for the face(s) **4, 5, 6** of additional blocks to be interconnected with the exposed faces **4, 5, 6** of any block **3** used to construct the wall. In one example, a user could build a floating staircase or floating shelf along the side of the vertical wall by interconnecting the faces **4, 5, 6** of half **2** or whole blocks **3** with the male **8** and female connector **7** features on the outer face **5** of the vertical wall.

[0070] The accompanying figures illustrate some of the numerous configurations and possible customizations of the block system of the present invention; however, it is to be understood that these figures are not exhaustive, and not all variations of this block system are included in these figures.

[0071] FIGS. 1A-B and 2A-B illustrate perspective views of embodiments of half blocks **2** wherein the male connectors **8** are cylindrical-shaped pegs and the female connectors **7** are square-shaped cavities. It is to be understood that the number and placement of male **8** and female **7** connectors may vary with different embodiments and block configurations. For example, the embodiment in FIG. 1A simply illustrates a single male connector **8** on one side face **6**, whereby FIG. 1B is the perspective view of FIG. 1A flipped vertically, illustrating two male connectors **8** on the inner face

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[0072] The configuration, placement, and number of male **8** and female connectors **7** may vary from half block **2** to half block **2**, as shown. FIGS. **3-6** illustrate whole blocks **3** constructed from these embodiments of half blocks **2** comprising peg-shaped male connectors **8** and square-shaped female connectors **7**. In an exemplary embodiment, a whole block **3** is constructed when two half blocks **2** are interconnected, preferably inner face **4** to inner face **4**, although it is possible for the half blocks to be interconnected inner face **4** to outer face **5**, and outer face **5** to outer face **5**. For example, when two half blocks **2** of FIG. **1A-B** are interconnected inner face **4** to inner face **4**, the resulting conformation is shown in FIG. **3**. By disconnecting the two half blocks **2**, rotating the half blocks **2** with respect to one another, and re-connecting as a whole block **3**, the orientation and placement of the male **8** and female connectors **7** may be easily changed, as shown at least in FIG. **4**.

[0073] FIG. **4** illustrates a whole block **3** cube with male connectors **8** on opposing sides **5**. If the half block **2** of FIG. **1A** is rotated 180 degrees and then interconnected (inner face **4** to inner face **4**) with the half block **2** of FIG. **1B**, the result would be the whole block **3** cube shown in FIG. **4** with male connectors **8** on two opposing outer faces **5**. Additionally, depending on the orientation and placement of the male **8** and female connectors **7** on the inner face **4**, a half block **2** (such as FIG. **2A**) can be rotated 90 degrees and interconnected with second half block **2** (such as FIG. **2B**) to create a whole block **3** cube with a male connector **8** on two adjacent outer faces **5** (block not shown). FIGS. **5** and **6** show alternative arrangements whereby the half block **2** of FIG. **1A** is flipped vertically and its outer face **5** is interconnected with the inner face **4** of the half block **2** of FIG. **1B**.

[0074] When making structures with conventional building blocks, the orientation of male **8** and female connectors **8** cannot be altered, and typically a building structure will inevitably have projecting male connection features exposed. For example, a rectangular vertical wall built from Lego®-type blocks will have male connector pegs projecting from the top of the wall unless capped off by a different type of block designed to hide the male pegs. The blocks **2, 3** of the present invention do not require a specific block designed to cover the male connector(s) **8**. Rather, the ability to reconfigure and orient the block(s) **2, 3** to hide the male connectors **8** is inherent in the design of the block system **1**, thereby creating a building structure with smooth, flat outer faces **5** (illustrated by FIG. **8**), if so desired.

[0075] FIG. **7** shows an embodiment of two whole blocks **3** with each block **3** having a plurality of male connectors **8** on a single outer face. The blocks are oriented with respect to one another such that the outer faces **5** comprising the male connectors **8** face one another for interconnection, while the remaining outer faces **5** only comprise female connectors **7**. FIG. **8** illustrates these two blocks **3** combined so that no male connectors **8** are showing. In other words, FIG. **8** illustrates a block structure **9** that is constructed when the blocks **3** of FIG. **7** are interconnected.

[0076] The block system **1** of the present invention may be created in a variation of shapes while still maintaining connectivity and the reconfigurability features of the present invention. Alternative embodiments are possible wherein the shape and design of male **8** and female connectors **7** may vary from block to block or within the same block. For example, a system of blocks can be designed where the blocks may vary in shape, but all of the blocks comprise the same shape of male **8** and female connectors **7** for interconnectivity among all the block faces **4, 5, 6**. However, the desire may exist to design a block wherein more than one shape of male **8** and female connector **7** may be included on the same block. It is to be understood that the shape of the block(s) and the design, shape, and number of the male **8** and female connectors **7** can vary while still maintaining the spirit and scope of the invention. The faces **4, 5, 6** of each half block **2** comprise a plurality of male **7** and female connectors **8** that may be oriented as desired and/or needed prior to interconnecting with the face **4, 5, 6** of another block.

[0077] FIGS. **9-29** illustrate alternative embodiments of the block system **1**, either through

variations in shape of the block **2**, **3** and/or variations in shape and design of the male **8** and/or female connectors **7**. These figures illustrate the versatility in the design of the blocks **2**, **3** and the connectors **7**, **8** while maintaining compatibility from one block **2**, **3** design to another.

[0078] For example, FIGS. **9-11** illustrate contemplated design variations such as a half-cylinder block shape (FIG. **9**), a triangular prism block shape (FIG. **10**), and a hexagonal block shape (FIG. **11**), ideal for creating rounded or sharp angled structures. The embodiment shown in FIG. **10** illustrates a half block **2** comprising two shape variations of the female connector **7** on the same half block **2**. In this embodiment, the female connectors **7** are round on the inner face **4** or outer face **5** and square-shaped on the side faces **6**; the corresponding male connectors **8** are cylindrical-shaped pegs dimensioned to fit within both shape variations of the female connector **7**.

[0079] Some embodiments of the female **7** and male connectors **8** are designed to provide a freedom of movement and interconnection versatility that is unique to this block system **1**. FIG. **11** shows the inner face **4** of hexagonal-shaped half block **2** with a female connector **7** that comprises a circular-shaped channel disposed in the center of the block **2** with an additional plurality of smaller circular-shaped female connectors **7** surrounding the channel-shaped connector **7**, similar to the female connectors **7** embodied in FIG. **10**. In this block **2**, **3** configuration, it is contemplated that a male-connector **8** can be inserted into the circular channel-shaped female connector **7** and slide circumferentially around the channel, thus allowing two half blocks **2** to be rotated 360 degrees with respect to one another. Other popular building block shapes may be employed as well, as it is contemplated that the block **2**, **3** may be shaped as desired and may include a plurality of female **7** and/or male **8** connection features on each face **4**, **5**, **6**.

[0080] FIGS. **12** through **29** illustrate alternative embodiments of the male **8** and female connectors **8** of a cube-shaped block **2**. The half block **2** shown in FIG. **12** includes a plurality of square-shaped female connectors **7** on an inner face **4** and two side faces **6**, adapted to accept the square-shaped male connectors **8** on the inner **4** and side faces **6**.

[0081] FIG. **13** illustrates a whole block **3**, wherein the female connector **7** disposed in the center of the outer face **5** comprises a square-shaped channel, and the female connector **7** on one or more of the side faces **6** comprises a squared-off U-shaped channel, such that when two half blocks **2** are conjoined, the female connector **7** of the side faces **6** may form a square-shaped female connector **7** that corresponds in size and shape to the female connector **7** shown on the outer face **5**. In this embodiment, the male connector **8** is preferably an L-shaped projection with dimensions and placement configured for insertion into the female connector **7**, although the male connector **8** may also be peg or square shaped (as illustrated in other embodiments) while maintaining compatibility with this embodiment of female connector **7**. A square or peg-shaped male connector **8** may be inserted into the channel of the female connector **7** and slide along the channel, allowing an offset in block interconnection and assembly.

[0082] FIG. **14** illustrates an embodiment of a half block **2** that, when combined with an identical half block **2**, creates the whole block **3** cube of FIG. **15**. FIGS. **14** and **15** show alternative design embodiments of the male **8** and female connectors **7**. Three female connectors **7** designed as square-shaped or T-shaped cavities are disposed on the side faces **6** around the periphery of a larger cross-shaped cavity centrally located on the inner face **4**, all functioning as female connector **7** features adapted to accept the male connectors **8** located on the inner **4** and side faces **6**. When the half blocks **2** of FIG. **14** are interconnected, the whole block **3** of FIG. **15** is formed, and the square-shaped female connectors **7** of the half block **2** align to form a cross-shaped female connector **7** on the outer faces **5** of the whole block **3**, thus illustrating how the shape of the female connector **7** may change when two half blocks **2** are interconnected.

[0083] In one or more embodiments, the block(s) **2**, **3** of the present block system **1** may be designed to be compatible with other popular brands of building blocks. FIG. **16** is a perspective view of two square prism half blocks **2** illustrating an embodiment of the male **8** and female connectors **8**, whereby the two half blocks **2** are aligned to be interconnected. The male connectors

8 are peg-shaped and sized to be compatible with Lego®-type blocks. The female connectors **7** are adapted to receive the peg-shaped male connectors of this block or other compatible blocks. While compatible with blocks such as Lego® or Mega Bloks®, the present blocks **2**, **3** do not have the design restrictions of the aforementioned brands as discussed previously herein. The blocks of the present invention include male **8** and female connectors **7** on all faces **4**, **5**, **6** of the block and may be reconfigured or re-oriented with respect to one another to allow for any desired combination. [0084] FIG. **17A** through FIG. **21** illustrate exemplary embodiments wherein more than one shape of male **8** and female connector **7** may be included on the same block. In FIGS. **17A-C**, the half blocks **2** comprise a peg shaped male connector **8** on the side faces **6** of the half block **2** and a plurality of square-shaped female connectors **7** on the side **6** and outer faces **5** of the half block **2**. However, the inner faces **4** of the half blocks **2** have male connectors **8** and corresponding female connectors **7** that differ in shape from the male **8** and female connectors **7** of the side **6** and outer faces **5**. In this way, one half block **2** may only be interconnected inner face **4** to inner face **4** with a second half block **2** to create a whole block (shown in FIGS. **18-21**); however, the any of the outer faces **5** of the whole block **3** may be interconnected with any corresponding outer faces **5** of any other embodiment of whole block **3**, such as those shown in at least FIGS. **1A** through FIG. **13**. There may be instances in which this design is preferred. Even though the half block **2** of this embodiment may only be interconnected inner face **4** to inner face **4**, certain flexibilities in block structure **9** construction are maintained because the half blocks **2** may still be rotated prior to interconnection with one another half block **2** to reconfigure the placement and orientation of the male **8** and female connectors **7** on the outer face **5** of the whole block **3**.

[0085] As previously described herein, a unique feature of this block system **1** is the capability to interconnect the blocks in a staggered or offset interconnection, as illustrated by the embodiments shown in FIGS. **22-29**. FIG. **22** shows two half blocks **2** comprising cylindrical-shaped male connectors **8** and square-shaped female connectors **7** disposed in an arrangement about the inner **4** (and/or outer **5**) face that allows the half blocks **2** to be aligned at an offset, in this example an approximate 45 degree offset, and interconnected, as shown in FIG. **23**.

[0086] FIGS. **24** and **25** illustrate a similar offset interconnection of the two half blocks **2** of FIG. **24**. In this embodiment, the female connector **7** comprises a circular-shaped channel, similar to the female connector **7** described with respect to FIG. **11**, with a rounded tab-shaped male connector **8** adapted to fit within the channel of the female connector **7** and slidable within that channel. In this way, not only may the half blocks **2** and whole blocks **3** be interconnected at an offset, each section of block **2,3** may rotate with respect to one another, thus allowing innumerable arrangements and precise positioning of the block pieces. It is to be noted that the male connectors **8** comprising cylindrical-shaped and square-shaped projections (or pegs), as illustrated in other embodiments described herein, are compatible with this embodiment of female connector **7**. This block embodiment may be inserted in a block structure **9** where rotational movement or offset interconnection is desired, further demonstrating the unique flexibility and compatibility of this block system **1**.

[0087] The block embodiments of FIGS. **26-29** may be constructed using straight pull injection molding for ease of manufacturing and further illustrate the advantages of the plurality, design, and position of the male **8** and female connectors **7** of the block system **1**. In the embodiments illustrated in FIGS. **26-29**, the female connector **7** comprises a channel or groove with a length that exceeds the length of the male connector **8**. Because the peg-shaped male connector **8** can be inserted into or slid along the groove-shaped female connector **7** at any point along the groove, the half blocks **2** may be connected at an offset or staggered arrangement from one another, thus forming an alternative and unconventional block shape not typically seen in prior art building blocks. Furthermore, the male connector **8** of this embodiment may be slidably inserted into the groove of the female connector **7**, thereby providing a slidable interconnection of one block **2, 3** to another block **2, 3**.

[0088] In the particular configuration shown in FIG. 27, the position and orientation of the half blocks 2 allow for male connectors 8 on a single outer face 5. The other five outer faces 5 of the cube block 3 include a plurality of female connectors 7, but no additional male connectors 8. By rotating the half blocks 2 of FIG. 26, the channel or groove of the female connector 7 may be truncated, as shown on the right side face 5 of FIG. 27. Alternatively, the female connector 7 on the side face 6 of one half block 2 may correspond with the female connector 7 on the side face 6 of a second half block 2, thereby extending the length of the female connector 7 along the outer face 5 of the whole block 3 (as shown on the left side outer face 5 of FIG. 27).

[0089] FIGS. 28 and 29 illustrate a perspective view of an embodiment of two half blocks 2 aligned for interconnection. In these embodiments, the outer face 5 includes two channel or groove-shaped female connectors, and the inner face preferably includes two male prong-like connectors 7. The side faces 6 of each half block 2 are shown to include one male connector 8 on one side face 6 and a plurality of female connectors 7 on all the side faces 6. It is to be understood that the number and placement of male 7 and female connectors 8 may vary with different embodiments and block configurations, this embodiment simply illustrates a single male connector 8 on one side face 6.

[0090] FIGS. 30-36 represent block structures 9 assembled from the blocks described herein, demonstrating the versatility of design and assembly. With conventional building blocks, the design of the block cannot be changed. However, the half blocks 2 of the present block system 1 can be reoriented with respect to one another, thereby altering the number and arrangement of male 8 and female interconnectors 7. This feature provides customization of the whole blocks 3 that can be created, thereby allowing for sturdier interconnection and more complex block structures 9.

[0091] FIGS. 30-31 illustrate that the blocks may be interlocked and assembled such that male connectors are hidden (FIG. 30) or projecting (FIG. 31) from the walls of the structure. In contrast to conventional building blocks such as Lego-type bricks, the blocks herein include male 8 and female connectors 7 on all faces 4, 5, 6 of the block 3 that may be reconfigured or re-oriented with respect to one another. A structure 9 built with the block system 1 herein may include male 8 and female connectors 8 on the sides of the wall (See FIGS. 30 and 31). This feature allows for the face(s) 4, 5, 6 of additional blocks to be interconnected with the exposed faces 4, 5, 6 of any block 3 used to construct the wall.

[0092] FIGS. 32-36 illustrate in exploded and assembled views how the blocks may be aligned for various structure assembly. FIGS. 32-34 illustrate a block structure whereby the male connectors 8 are hidden from the outer faces 5 of the block structure 1. FIGS. 35-36 illustrate an alternative arrangement whereby a structure 1 may be assembled to include male connectors 8 on the outer faces 5.

[0093] FIG. 33 is an example of a structure 9 built with the blocks 3 having male connectors 8 on only one side, similar to conventional building blocks. The limitation of this conventional block is that complex, sturdy structures cannot be produced. The upper right side of FIG. 33 shows that there is no connection between two adjacent blocks, which causes the assembly to be less stable. This effect would be worsened on even larger, more complicated structures. FIG. 32 shows the advantages of the reconfigurable and customizable block system 1 herein by allowing a block to be reoriented into the configuration shown in FIG. 4. This allows the gap to be filled with a connection, thereby creating a sturdier assembly.

[0094] In certain embodiments, half blocks may be produced in various colors, transparencies, or materials to enhance the visual appeal of an assembly. For example, half blocks can be manufactured with integrated lighting or formed from injection-molded plastic, though alternative materials and manufacturing methods may be employed as warranted. Notably, half blocks remain compatible and connectable regardless of color, transparency, or material differences.

[0095] In addition, accessory blocks of different shapes from the half blocks may be utilized. While these accessory blocks may not have the same functionality as the half blocks, they include compatible connectors enabling them to interlock with the half blocks. Such accessory blocks may

range from simple, single-piece parts to more complex elements and assemblies (e.g., pistons, motors, hinges, rotators, sliders, controllers, and the like).

[0096] Although the invention has been explained in relation to its preferred embodiment(s) as mentioned above, it is to be understood that many other possible modifications and variations can be made without departing from the scope of the present invention. It is, therefore, contemplated that the appended claim or claims will cover such modifications and variations that fall within the true scope of the invention.

[0097] Alternative language and synonyms may be used for any one or more of the terms discussed herein. No special significance is to be placed upon whether or not a term is elaborated or discussed herein. Synonyms for certain terms are provided. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification, including examples of any terms discussed herein, is illustrative only, and is not intended to further limit the scope and meaning of the disclosure or of any exemplified term. Likewise, the disclosure is not limited to various embodiments given in this specification.

[0098] It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element, without departing from the scope of the present inventive subject matter. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

[0099] Embodiments of the inventive subject matter are described herein with reference to plan and perspective illustrations that are schematic illustrations of idealized embodiments of the inventive subject matter. As such, variations from the shapes of the illustrations as a result, for example, of manufacturing techniques and/or tolerances, are to be expected. Thus, the inventive subject matter should not be construed as limited to the particular shapes of objects illustrated herein, but should include deviations in shapes that result, for example, from manufacturing. Thus, the objects illustrated in the figures are schematic in nature and their shapes are not intended to illustrate the actual shape of a region of a device and are not intended to limit the scope of the inventive subject matter.

[0100] In the drawings and specification, there have been disclosed typical preferred embodiments of the inventive subject matter and, although specific terms are employed, they are used in a generic and descriptive sense only and not for purposes of limitation, the scope of the inventive subject matter being set forth in the following claims.

Claims

1. A reconfigurable and customizable block system comprising: a half block comprising: an inner face, an outer face, and a plurality of side faces; wherein the inner face has at least one male connector and at least one female connector; wherein at least one of the side faces includes at least one female connector and at least one of the side faces includes at least one male connector; wherein the outer face has at least one female connector or at least one male connector; whereby the at least one male connector of a first half block can be interconnected with the at least one female connector of a second half block, thereby creating a block structure.
2. The reconfigurable and customizable block system of claim 1, whereby a whole block is formed when the at least one male connector of the inner face or outer face of a first half block interconnects with the at least one female connector of the inner face or outer face of a second half block, the whole block comprising: a plurality of outer faces; and whereby the orientation of the side faces of the first half block when interconnected with the second half block determines the number and orientation of male connectors and female connectors on the outer faces of the whole block.

3. The reconfigurable and customizable block system of claim 1, whereby the two half blocks are the same shape.
4. The reconfigurable and customizable block system of claim 1, whereby the side faces of the half block are rectangularly shaped.
5. The reconfigurable and customizable block system of claim 1, whereby the at least one female connector has a length that exceeds a width of the at least one male connector, whereby the male connector may be interconnected at any point along the length of the female connector, thereby allowing an offset interconnection.
6. The reconfigurable and customizable block system of claim 1, whereby the first half block and the second half block are configured for rotation with respect to one another.
7. The reconfigurable and customizable block system of claim 1, whereby the half block is designed in such a way to allow for straight pull injection molding.
8. The reconfigurable and customizable block system of claim 1, whereby the inner face of a first half block can be interconnected with the inner face or outer face of a second half block at 45 degrees.
9. The reconfigurable and customizable block system of claim 1, whereby the half block comprises a plurality of female connectors, and whereby at least one female connector is a first shape and at least one female connector is a second shape.
10. The reconfigurable and customizable block system of claim 1, whereby the half block comprises a plurality of male connectors, and whereby at least one male connector is a first shape and at least one male connector is a second shape.
11. The reconfigurable and customizable block system of claim 1, whereby the at least one male connector and at least one female connector occupy less than 25 percent of a surface area of the outer face and the side face.
12. A reconfigurable and customizable block system comprising: a whole block comprising: two half blocks, each half block comprising: an inner face, an outer face, and a plurality of side faces; whereby the inner face comprises at least one connecting feature configured to interconnect to the inner face of a similarly shaped half block; whereby at least one of the side faces comprises at least one connecting feature configured to interconnect to the side face and the outer face of a half block; whereby the interconnection of the inner faces of the two half blocks creates a whole block; and whereby the orientation of the connector features on the outer faces and side faces can be reoriented by disconnecting the two half blocks from one another, rotating the half blocks with respect to one another, and re-connecting the inner faces of the two half blocks to create a whole block with an altered configuration of connector features.
13. The reconfigurable and customizable block system of claim 12, whereby the connecting feature of the inner face can be reoriented for interconnection to the side face, the inner face, and the outer face of a half block.
14. The reconfigurable and customizable block system of claim 12, whereby the alignment of two side faces creates a square face.
15. The reconfigurable and customizable block system of claim 12, whereby the connector features comprise a male connector and a female connector, and a whole block is formed when the at least one male connector of the inner face of a first half block interconnects with the at least one female connector of the inner face of a second half block, the whole block comprising: a plurality of outer faces, wherein each outer face includes at least one female connector, and wherein at least one outer face includes at least one male connector; and whereby the orientation of the side faces of the first half block when interconnected with the second half block determines the number and orientation of male connectors on the outer faces of the whole block.
16. The reconfigurable and customizable block system of claim 12, whereby the two half blocks are the same shape.
17. The reconfigurable and customizable block system of claim 12, whereby the at least one male

connector and at least one female connector occupy less than 25 percent of a surface area of the outer face and the side face.

18. The reconfigurable and customizable block system of claim 12, whereby the whole block can have only one outer face with male connectors in a first orientation, or two outer faces with male connectors in different orientations.

19. The reconfigurable and customizable block system of claim 12, whereby the half block is designed in such a way to allow for straight pull injection molding.

20. A reconfigurable and customizable block system comprising: a whole block comprising: two half blocks, each of the half blocks comprising: an inner face, an outer face, and a plurality of side faces; whereby the inner face has at least one male connector and a plurality of female connectors; whereby at least one side face has at least one male connector; whereby the outer face and side face has a plurality of female connectors; whereby the male connector and female connector are adapted to interconnect via interference fit; whereby interconnection of the inner faces of the two half blocks creates the whole block, the whole block having six outer faces; and whereby the number of outer faces comprising male connectors can be changed by re-orienting the two half blocks with respect to one another.
